M. ADITYA PATRA

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Summary

B.Tech Computer Science student building end-to-end machine learning systems on real public data, from ingestion and statistical validation to explainable models, dashboards and tested pipelines. Recent work includes an orbital collision-screening engine that processes 14,000+ satellites and debris objects, and a Formula 1 strategy platform built on the full 2024 season. Focused on honest evaluation: leakage checks, group-aware splits, and auditing where a model's accuracy really comes from. Seeking an ML Engineer or Data Science internship.

Technical Skills

Languages	Python, SQL (MySQL), Java, C
ML & Data	scikit-learn, XGBoost, TensorFlow/Keras, SHAP, pandas, NumPy, SciPy, imbalanced-learn (SMOTE), lifelines
Methods	Feature engineering, cross-validation (stratified, group-aware), leakage prevention, hypothesis testing (Welch t-test, Chi-square), survival analysis (Kaplan–Meier, Cox PH), clustering (K-Means, DBSCAN)
Apps & Viz	Plotly Dash, Streamlit, Plotly, Matplotlib, Seaborn
Engineering	Git, GitHub, pytest, Jupyter, VS Code, MySQL Workbench

Experience

Data Analyst & Machine Learning Intern June 2026 – August 2026
CloudExperts.co *Remote*
- Reviewed and analysed production codebases for a retrieval-augmented generation (RAG) web application built on React/TypeScript and AWS Lambda.
- Gained hands-on exposure to cloud architecture and worked on stock market analysis.

Logistics & Organizing Member 2023 – Present
Concordia Club, DY Patil International University *Pune*
- Coordinated logistics and participant management for 3+ large university events, including Model United Nations, with 100+ participants across concurrent sessions.

Projects

OrbitGuard: Orbital Collision Screening & Risk | *Python, SGP4, NumPy, SciPy, pytest* Oct 2026 – Present
`https://github.com/aditya-patra1011/Orbitguard`
- Built a conjunction-screening engine that pulls live CelesTrak orbital data, propagates every object with SGP4 and finds close approaches between all pairs; screened **14,391 objects over 24 h in 273 s**, finding 47,753 conjunctions under 5 km.
- Designed a KD-tree search with a velocity-padded radius so fast head-on encounters cannot be missed (covered by a unit test), plus a linear relative-motion pre-filter that made the pipeline ~**6× faster** with identical results.
- Dropped stale element sets and flagged co-orbital formation flyers to keep results meaningful; next phases add collision probability, an ML risk model on ESA Kelvins data, and a FastAPI service.

Kepler Exoplanet Classification | *Python, scikit-learn, XGBoost, SMOTE, Streamlit* Apr 2026 – May 2026
`https://github.com/aditya-patra1011/kepler-exoplanet-classification`
- Classified 9,564 Kepler objects as confirmed planets or false positives; Random Forest reached **F1 0.9943, ROC-AUC 0.9989** on a 1,241-object held-out set, ahead of XGBoost and Gradient Boosting.
- Audited the result: with NASA's vetting flags removed, F1 drops to about 0.94, which is the honest measure of what the model learns from raw physical measurements.
- Handled class imbalance with SMOTE on training data only, confirmed feature separation with Welch t-tests ($p < 0.001$), and shipped an interactive Streamlit dashboard.

Exoplanet Candidate Classification & Habitability | *Python, XGBoost, TensorFlow, SHAP, Streamlit* Sep 2026 – Oct 2026
`https://github.com/aditya-patra1011/exoplanet-habitability-project`
- Classified Kepler's unresolved candidate signals using six models (RF, XGBoost, GBM, SVM, LR, MLP) on 7,587 labelled objects, with 5-fold group-aware cross-validation across three feature tiers to separate physical signal from vetting-flag leakage.
- Scored confirmed and predicted-real planets for habitable-zone status (Kopparapu insolation bounds, rocky-radius cut) and a simplified Earth Similarity Index; **3.21%** fall in the conservative habitable zone, and known planets such as TRAPPIST-1e and Kepler-186f were recovered.

BoxBox: F1 Strategy Intelligence Platform | *Python, FastF1, scikit-learn, lifelines, SHAP, Dash* Aug 2026 – Oct 2026
`https://github.com/aditya-patra1011/boxbox`$_{s}$*trategy*$_{i}$*ntelligence*
- Built a 9-phase pipeline on the full 2024 F1 season (23 circuits): data collection, cleaning, tyre-degradation modelling, survival analysis, strategy simulation and an interactive 5-page Plotly Dash dashboard.

- Trained per-circuit/compound degradation models selected by cross-validated RMSE (Random Forest won 39 of 59, Ridge 20) and six purpose-built models covering stint length, compound choice, pit window and undercut viability, explained with SHAP.
- Used Kaplan–Meier and Cox proportional-hazards models to find circuit-specific tyre risk drivers; validated manual data cleaning independently with DBSCAN, and found and fixed two upstream bugs that had been flattening risk scores.

Football Archive Relational Database | *MySQL* Dec 2025 – Jan 2026
`https://github.com/aditya-patra1011/football-archive`

- Designed a normalised (3NF) MySQL database of 11 tables covering the Premier League, La Liga and Bundesliga, enforced with 16 foreign keys and STRICT mode.
- Built 5 analytical views, 3 stored procedures and 12 queries, including xG over/under-performance, goals per 90 and clean-sheet rates.

Education

B.Tech in Computer Science & Engineering 2024 – 2028
DY Patil International University, Pune
Class 12 2023
Army Public School, Bhopal

Certifications

Data Mining in Python, University of Michigan (2024) | Technology Commercialization, University of Rochester (2024)

Languages

English (Fluent), Hindi (Fluent), Odia (Native), French (Basic)